ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

SYSTEMS ANALYSIS AND DESIGN

PRACTICE EXAM · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Systems Analysis and Design practice exam.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training marking guideline.

How to mark: each ✓ is one mark. Answers must relate to the case study where asked. Accept either Gane and Sarson or Yourdon DFD notation and any standard ERD notation (crow's foot or Chen). Accept other correct answers.

SECTION A

QUESTION 1

1.1 1.1.1 D ✓ 1.1.2 B ✓ 1.1.3 D ✓ 1.1.4 B ✓ 1.1.5 C ✓ 1.1.6 A ✓ 1.1.7 B ✓ 1.1.8 D ✓ 1.1.9 A ✓ 1.1.10 C ✓ (10)

1.2 1.2.1 False ✓ 1.2.2 True ✓ 1.2.3 True ✓ 1.2.4 True ✓ 1.2.5 False ✓ 1.2.6 True ✓ 1.2.7 False ✓ 1.2.8 True ✓ 1.2.9 True ✓ 1.2.10 False ✓ (10)

1.3 1.3.1 I ✓ 1.3.2 H ✓ 1.3.3 J ✓ 1.3.4 F ✓ 1.3.5 B ✓ 1.3.6 D ✓ 1.3.7 A ✓ 1.3.8 C ✓ 1.3.9 G ✓ 1.3.10 E ✓ (10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2

2.1 Any THREE: books get lost because nobody knows who has them ✓; finding overdue books takes hours ✓; records in a book are slow to search / can be damaged ✓; fines are calculated by hand (errors); no reminders are sent to members (3)

2.2 Any TWO SMART-style objectives, e.g. record every loan and return electronically ✓; send SMS reminders automatically when a book is overdue ✓; calculate fines automatically; produce a list of overdue books in seconds (2)

2.3 Technical ✓ – is the hardware, software and skills available (e.g. one PC, library software, an SMS service) ✓; economic ✓ – do benefits outweigh costs within the small municipal budget (saved staff time, fewer lost books) ✓; operational ✓ – will the librarian and members accept and use the system / training needed ✓; schedule ✓ – can it be developed and implemented in the time available (e.g. before the new year) ✓ (8)

2.4 Parallel conversion ✓ – keep the loans book for a while so no records are lost if problems occur (accept phased / direct with a valid reason, e.g. direct because the library is small and cheap to convert) ✓ (2)

[15]

QUESTION 3

3.1 (a) Interview ✓ – one person, detailed answers and follow-up questions possible ✓ (b) Questionnaire ✓ – many members can be reached quickly and cheaply ✓ (4)

3.2 Any THREE relevant questions ✓✓✓, e.g. How many books are borrowed per day? How do you currently find overdue books? What reports would you like? How are fines recorded? What problems do you experience most? (3)

3.3 Any ONE: people may act differently when watched ✓; time-consuming; may not see all situations (e.g. month-end) (1)

3.4 Any THREE: users see and try the system early ✓; misunderstandings and missing requirements are found before coding is finished ✓; users feel involved and accept the system more easily ✓; changes are cheaper early in the project (3)

[11]

QUESTION 4

4.1 One central process "Library Management System" ✓; external entities, e.g. MEMBER ✓, LIBRARIAN, SMS PROVIDER / MUNICIPALITY (reports); any FOUR correctly labelled flows with correct direction ✓✓✓✓, e.g. member details →, book request →, ← loan confirmation, ← SMS reminder / overdue notice, fine payment →, ← receipt, overdue report → librarian; neat, correct symbols, no data stores in a context diagram ✓✓ (8)

4.2 External entity MEMBER ✓; process 1 "Record return" ✓ – updates LOANS data store ✓ (return date) and BOOKS data store (status available) ✓; process 2 "Calculate fine" ✓ – reads date due from LOANS ✓, writes to FINES data store ✓; flow "fine amount / receipt" back to MEMBER ✓ (8)

4.3 External entity ✓; process ✓; data store ✓; data flow ✓ (4)

[20]

QUESTION 5

5.1 2 marks per entity (key ✓, two attributes ✓): MEMBER – MemberNo; Name, IDNumber, Address, Cellphone; BOOK – BookNo (or ISBN + copy number); Title, Author, Status; LOAN – LoanNo; BorrowDate, DueDate, ReturnDate (plus MemberNo, BookNo as foreign keys); FINE – FineNo; Amount, DatePaid (6)

5.2 Three entities drawn correctly ✓; MEMBER one-to-many LOAN ✓✓ (one member has many loans); BOOK one-to-many LOAN ✓✓ (a book is borrowed many times over time). Accept a many-to-many MEMBER–BOOK resolved by LOAN. (5)

5.3 Field name: MemberCellphone ✓; data type: text / string (not number, to keep the leading 0) ✓; size: 10 characters ✓; format: 0XX XXX XXXX (10 digits) ✓; validation: length check – exactly 10 digits and starts with 0 ✓ (5)

5.4 Any FOUR: logical order of fields (member, then book, then dates) ✓; clear labels ✓; default values (today's date, due date calculated automatically) ✓; drop-down lists / look-ups to reduce typing errors ✓; consistent layout and colours; error messages that explain how to fix the problem; Save and Cancel buttons (4)

5.5 Example: "Dear [Name], the book [Title] was due on [DueDate]. Please return it. Fine so far: R[amount]. Ubuntu Community Library" ✓ – fields in brackets come from the MEMBER, BOOK and LOAN tables ✓ (2)

5.6 $17 - 3 = 14$ days late ✓ $\times$ R2 = R28 ✓ (2)

[24]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE